Emotion Analysis Using BRIGHTER + EthioEmo Datasets

A Comparative Analysis of Multilingual Transformer Models for Multi-Label Emotion Classification in Bantu Languages

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1. Introduction

Emotion analysis plays a vital role in Natural Language Processing (NLP), with applications in areas such as social media monitoring, public opinion analysis, and mental health assessment. However, research has predominantly focused on high-resource languages like English, leaving African languages significantly underrepresented despite Africa's linguistic diversity of over 2,000 languages. The limited availability of high-quality annotated datasets has hindered the development of effective NLP systems for these languages. The BRIGHTER and EthioEmo datasets offer a promising solution by providing multilingual, multi-label emotion annotations. This project aims to evaluate the performance of multilingual transformer models, specifically mBERT and XLM-R, for multi-label emotion classification in linguistically related Bantu languages, ultimately providing insights into cross-lingual transfer learning in low-resource contexts.

2. Background and Literature Review

Multilingual transformer models such as mBERT and XLM-R have significantly advanced cross-lingual NLP, with XLM-R often outperforming mBERT due to its larger and more diverse pretraining corpus (Conneau et al., 2020). However, performance on African languages remains uneven, hindered by high typological diversity and a scarcity of annotated training data (Adelani et al., 2022; Muhammad et al., 2023). The BRIGHTER dataset (Abdulmumin et al., 2024) addresses this resource gap by providing multi-label emotion annotations across 32 languages, including numerous African varieties, while EthioEmo supplements Ethiopian language coverage. Prior work has largely emphasized broad cross-continental transfer, overlooking the potential benefits of leveraging intra-family linguistic relatedness. This project addresses that oversight by restricting evaluation to a controlled set of Bantu languages (isiZulu, isiXhosa, and Swahili), thereby enabling a rigorous analysis of how genealogical similarity influences the fine-tuning efficacy of mBERT and XLM-R in low-resource, multi-label emotion classification.

3. Research Questions

This project aims to answer the following questions:

1. How do **mBERT** and **XLM-R** compare in multi-label emotion classification for Bantu languages?
2. To what extent does linguistic similarity influence model performance across **isiZulu, isiXhosa,** and **Swahili**?
3. What types of errors do these models make in multi-label emotion classification tasks?

Understanding these factors is important for improving NLP systems in low-resource settings and guiding the selection of appropriate models for African language applications.

4. Dataset and Data Description

The primary dataset used in this project is a combination of the BRIGHTER and EthioEmo datasets for multi-label emotion classification.

The datasets consist of short text samples annotated with multiple emotion labels, including anger, fear, joy, sadness, and surprise. Each instance is represented with a binary label vector (i.e. $[0, 1, 1, 0, 1]$) to support multi-label classification. The data is sourced from multiple domains such as social media and curated text, and annotations are thankfully provided by fluent speakers, improving label reliability.

Dataset sizes vary across language, with each containing several thousand samples, which is sufficient for fine-tuning pretrained models. Error analysis is conducted to examine misclassification and emotion confusion, which would provide greater insight into model behaviour and the influence of linguistic similarity across languages.

5. Approach and Methodology

The study evaluates multi-label emotion classification in bantu languages (isiZulu, isiXhosa, and Swahili) using a comparative experimental framework. A traditional baseline model such as Logistic regression with TF-IDF features (Das et al. #) is implemented alongside multilingual transformer models, specifically mBERT and XLM-R, to assess the effectiveness of transfer learning in low resource settings.

The task is formulated as a multi-label classification problem using sigmoid activation and binary cross-entropy loss, reflecting the presence of multiple emotions per text instance. Standard preprocessing, including text normalisation, model-specific tokenisation, and binary label encoding, will be applied consistently across all models.

All models are trained and evaluated under controlled conditions using identical data splits , evaluation metrics, as well as comparable hyperparameter tuning to ensure fair comparison across languages.

6. Evaluation Strategy

Model performance will be evaluated using standard multi-label classification metrics, with F1-score as the primary metric due to its robustness against class imbalance. Alongside the F1 score, loss, accuracy, precision, and recall will be used as supplementary metrics.

A traditional logistic regression model will serve as a baseline, with performance compared against mBERT and XLM-R to assess the effectiveness of transformer-based approaches. Evaluation will also include cross-lingual analysis by comparing performance across the defined languages, enabling assessment of model consistency and the impact of linguistic similarity. The success of a model will be measured by the improvements in the F1 score over baseline (before performing any type of training then after full fine-tuning) and consistent performance across languages.

7. Expected Outputs and Contributions

The project sets out to produce a comparative evaluation of multilingual transformer models for multi-label classification in Bantu languages, providing insights into how linguistic similarity influences model performance across these languages. Identifying common misclassifications patterns and emotion contribution trends would aid in better understanding of model behaviour in low-resource environments.

These findings will provide evidence for underrepresented African languages and provide practical guidance for selecting appropriate models in multilingual NLP applications. The potential challenges include class imbalance, limited dataset sizes and computational constraints.

8. Feasibility, Timeline & Work Plan

The project is designed to be feasible within a five-week timeframe due to the use of publicly available datasets and pretrained multilingual models such as mBERT and XLM-R, which reduce training time and computational requirements. The controlled scope, focusing on a limited number of models and languages, further supports timely completion.

Timeline:

- **Week 1:** Literature review and dataset preparation
- **Week 2:** Baseline implementation and initial experiments
- **Week 3:** Transformer fine-tuning (mBERT, XLM-R)
- **Week 4:** Evaluation, cross-lingual analysis, and error analysis
- **Week 5:** Report writing and finalisation

Potential risks include class imbalance and limited dataset sizes, which may affect model performance; however, these are mitigated through appropriate evaluation metrics and controlled experimental design and data augmentation techniques.

9. References

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